Network-based multi-omics disease-drug associations reveal drug repurposing candidates for COVID-19 disease phases

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Introduction

The development and roll-out of vaccines, and the use of various drugs have contributed to controlling the COVID-19 pandemic. Nevertheless, challenges such as the unequal vaccine distribution, emerging variants, and weakened immune systems in some groups necessitate further drug development to combat the disease.

Objective

This study aimed to identify and prioritize drugs for mild, moderate, or severe COVID-19 using a network-based approach integrating curated drug and multi-omics data.

Methodology

This study aimed to identify and prioritize drugs for mild, moderate, or severe COVID-19 using a network-based approach integrating curated drug and multi-omics data.

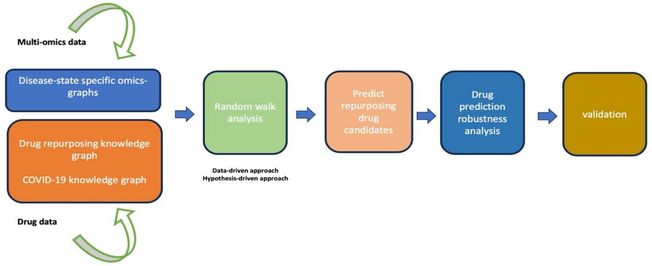

Figure 1. Diagram illustrating the workflow implemented in this study. The workflow begins with curating multi-omics data and drug data followed by data pre-processing, a random walk with restart network analysis using both data-driven and hypothesis-driven approaches. Next, we prioritized and characterized candidate drugs, followed by drug prediction robustness analysis. Finally, we concluded the analysis by validating the predicted drug candidates.

Table 1. Description of the node-types in drug repurposing knowledge graph

Node-type	Number of features
Anatomy	400
Anatomical Therapeutic Chemical (Atc)	4048
Biological Process	11381
Cellular Component	1391
Compound	24313
Disease	5103
Gene	39220
Molecular Function	2884
Pathway	1822
Pharmacologic Class	345
Side Effect	5701
Symptom	415
Taxonomy (Tax)	215

Table 2. Description of the node-types in COVID-19 Knowledge Graph

Node-type	Number of features
SARS-CoV-2 baits	23
Host genes and drug targets	10959
Pathways	274
Drugs (chemical/compound)	4266
Biological process (Phenotypes)	1893

Results

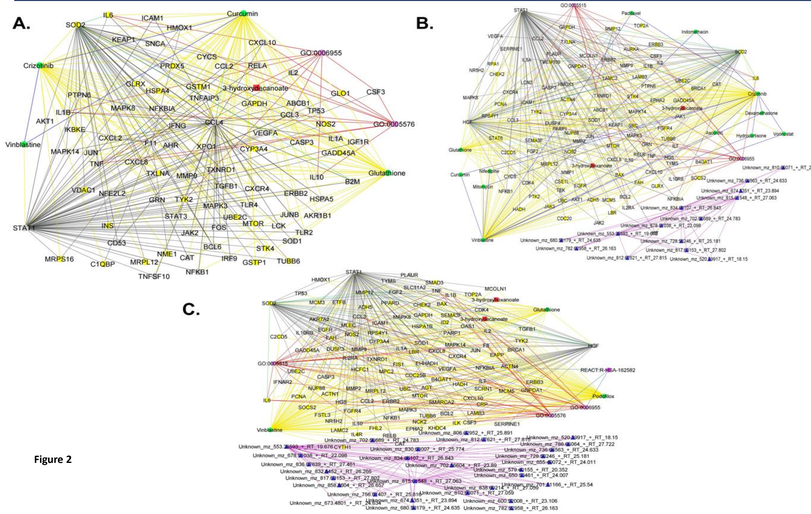

Figure 2

Figure 2 (A) Graph representation of the interaction between drugs (green nodes), proteins (yellow nodes), transcripts (grey nodes), metabolites (red nodes), lipids (blue nodes), and biological processes and pathways (pink nodes) as observed from predicting candidate drugs using existing knowledge graphs, mild disease-state specific omics-graphs, and data-driven seeds. The graph reveals distinct subnetworks formed by hub CCL4, demonstrating extensive interactions with drug candidates and other molecular features including seed nodes (STAT1 and SOD2)

(B) Graph representation of the interaction between drugs (green nodes), proteins (yellow nodes), transcripts (grey nodes), metabolites (red nodes), lipids (blue nodes), and biological processes and pathways (pink nodes) as observed from predicting candidate drugs using existing knowledge graphs, moderate disease-state specific omics-graphs, and data-driven seeds. The graph reveals distinct subnetworks formed by hub HGF, demonstrating extensive interactions with drug candidates and other molecular features including seed nodes (STAT1 and SOD2), as well as a subnetwork formed among lipids.

(C) Graph representation of the interaction between drugs (green nodes), proteins (yellow nodes), transcripts (grey nodes), metabolites (red nodes), lipids (blue nodes), and biological processes and pathways (pink nodes) as observed from predicting candidate drugs using existing knowledge graphs, severe disease-state specific omics-graphs, and data-driven seeds. The graph reveals distinct subnetworks formed by hub HGF, demonstrating extensive interactions with drug candidates and other molecular features including seed nodes (STAT1 and SOD2), as well as a subnetwork formed among lipids.

Yellow edges represent drug-protein and drug-transcript pairwise interactions. Red edges represent biological process-protein interactions and biological process-transcripts interactions. Pink edges represent pairwise interactions between lipids. Green edges represent protein-protein interactions. Black edges represent transcript-transcript interactions and protein-transcript interactions. Dark blue edges represent drug-drug interactions. Light blue edges represent biological processes-biological process interactions and biological process-pathway interactions.

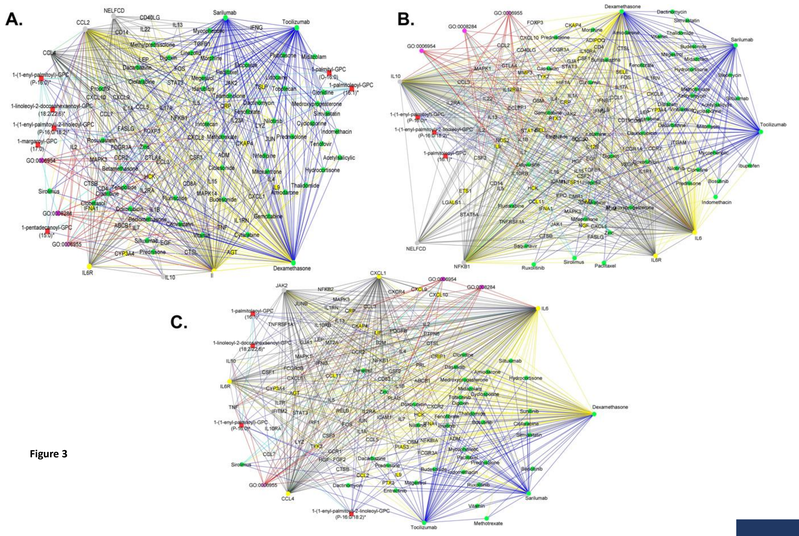

Figure 3

Figure 3. (A) Graph representation of the interaction between drugs (green nodes), proteins (yellow nodes), transcripts (grey nodes), metabolites (red nodes), and biological process (pink nodes) as observed from predicting candidate drugs using existing knowledge graphs, mild disease-state specific omics-graphs, and hypothesis-driven seeds. The graph reveals distinct subnetworks formed by hubs CCL2, CCL4, and NELFCD demonstrating extensive interactions with drug candidates, seed nodes (IL6 and IL6R), and other molecular features

(B) Graph representation of the interaction between drugs (green nodes), proteins (yellow nodes), transcripts (grey nodes), metabolites (red nodes), and biological process (pink nodes) as observed from predicting candidate drugs using existing knowledge graphs, moderate disease-state specific omics-graphs, and hypothesis-driven seeds. The graph reveals distinct subnetworks formed by hubs NFKB1, IL10, and NELFCD demonstrating extensive interactions with drug candidates, seed nodes (IL6 and IL6R), and other molecular features

(C) Graph representation of the interaction between drugs (green nodes), proteins (yellow nodes), transcripts (grey nodes), metabolites (red nodes), and biological process (pink nodes) as observed from predicting candidate drugs using existing knowledge graphs, severe disease-state specific omics-graphs, and hypothesis-driven seeds. The graph reveals distinct subnetworks formed by hubs CXCL1, CCL4, and JAK2 demonstrating extensive interactions with drug candidates, seed nodes (IL6 and IL6R), and other molecular features.

Yellow edges represent drug-protein and drug-transcript pairwise interactions. Red edges represent biological process-protein interactions and biological process-transcripts interactions. Green edges represent protein-protein interactions. Black edges represent transcript-transcript interactions and protein-transcript interactions. Blue edges represent drug-drug interactions. Light blue edges represent biological processes-biological process interactions and biological process-pathway interactions.

Table 3. Some potential drugs for mild, moderate, and severe COVID-19, ranked according to their measure of proximity to STAT1, SOD2, 3-hydroxyoctanoate, and unknown_mz_815.61548_+_RT_27.063 seed nodes as determined through RWR analysis of COVID-KG, DRKG, and DSOG.

Drug	Drug category	Mechanism of action potentially linked with COVID-19	Measure of proximity in mild	Measure of proximity in moderate	Measure of proximity in severe
Glutathione	Antioxidant	Protect cells from damage caused by oxidative stress	0.0007061	0.0007044	0.0007044
Crizotinib	Tyrosine kinase inhibitor	Express immune modulatory properties and could inhibit receptor tyrosine kinases and affect cellular processes relevant to viral replication	0.0007022	0.0007005	0.0007002
Curcumin	Natural compound	Expresses antiviral properties, immune modulatory and antioxidant properties that may contribute to inhibit inflammation, oxidative stress, and reduce lung injury	0.0006966	0.0006948	0.0006951
Vorinostat	Histone deacetylase inhibitor	Express anti-inflammatory and antiviral properties	0.0006943	0.0006952	0.0006954
Vinblastine	Anticancer	Disrupt microtubule dynamics, leading to mitotic spindle dysfunction and cell cycle arrest.	0.0006920	0.0007029	0.0007022

Table 4. Node degree, betweenness, and closeness centrality measures for the drug repurposing candidates predicted using the data-driven approach.

Drug name	Degree mild	Degree moderate	Degree severe	Betweenness mild	Betweenness moderate	Betweenness severe	Closeness mild	Closeness moderate	Closeness severe
Glutathione	212	234	118	0.0397	0.0350	0.0439	0.3532	0.3611	0.3657
Crizotinib	1919	1947	2	0.5147	0.4171	0.0025	0.5478	0.5297	0.3987
Curcumin	1076	4	2	0.3281	0.0005	0.0	0.5389	0.4317	0.3983
Vorinostat	2	4	2	0.0009	0.0005	0.0	0.4199	0.4317	0.3983
Vinblastine	4	1905	1378	0.0009	0.4787	0.6440	0.4364	0.5639	0.5556

Table 5. Some potential drugs for mild, moderate, and severe COVID-19, ranked according to their measure of proximity to IL6 and IL6R seed nodes as determined through RWR analysis of COVID-KG, DRKG, and DSOG. The references point to publications that have reported the drugs' mechanism of action potentially linked with COVID-19.

Drug	Drug category	Mechanism of action potentially linked with COVID-19	Measure of proximity in mild	Measure of proximity in moderate	Measure of proximity in severe
Dexamethasone	Corticosteroid	Suppresses immune response	0.0033655	0.0033691	0.0033697
Sarilumab*	Anti-interleukin 6 (IL6) receptor monoclonal antibody	Suppresses immune response by blocking IL6 signalling	0.0031924	0.0031908	0.0031911
Tocilizumab*	Anti-interleukin 6 (IL6) receptor monoclonal antibody	Suppresses immune response by blocking IL6 signalling	0.0031908	0.0031894	0.0031897
Zinc	Essential mineral / Nutrient	Expresses antiviral properties and interferes with viral RNA synthesis to inhibit replication	0.0002730	0.0002681	0.0002692
Sirolimus	Immunosuppressive drug	Expresses immunomodulatory and anti-inflammatory properties and inhibits the expression of proinflammatory cytokines.	0.0001524	0.0001526	0.0001530

Table 6. Node degree, betweenness, and closeness centrality measures for the drug repurposing candidates predicted using the hypothesis-driven approach.

Drug name	Degree mild	Degree moderate	Degree severe	Betweenness mild	Betweenness moderate	Betweenness severe	Closeness mild	Closeness moderate	Closeness severe
Dexamethasone	1633	1632	1633	0.6282	0.5881	0.6293	0.6355	0.6233	0.6228
Tocilizumab	830	829	830	0.11436	0.1008	0.1103	0.4643	0.4482	0.4571
Sarilumab	469	469	469	0.0286	0.0246	0.0283	0.4432	0.4289	0.4373
Sirolimus	7	8	6	0.0063	0.0068	0.0054	0.4831	0.4889	0.4512
Vitamin C	6	6	4	0.0047	0.0034	0.0004	0.4860	0.4525	0.4460

References

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Acknowledgment

Conclusion

This study demonstrates that the multi-omics data-driven integrative analysis using curated drug data enables prioritizing drug candidates for COVID-19 disease phases, offering a comprehensive basis for therapeutic strategies that can be brought to market quickly given their established safety profiles.

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